BT201 FINAL TERM MCQS AND SUBJECTIVE WITH MIDTERM CURRENT MCQS AND SUBJECTIVE

1. On Island and habitat patches on continents __**Species richness increases with area and decreases with isolation**
2. Succession in the intertidal zone require_____**1 to 3 years**
3. Highest rate terrestrial primary production occur under___**Warm and Moist Conditions**
4. _____ in physical and chemical conditions across aquatic and terrestrial environment can account for significant portion of the diversity among planktonic algae and terrestrial plants__**Heterogeneity**
5. The refuges that promote the persistence of hosts and prey include____ **Secure places to which the exploiter has limited access**
6. The environment Limits population growth by____ **Changing Birth and Death rates**
7. An environment in which resources are patchy also fosters____ **Clumped Distribution**
8. Eusociality is_____ **More complex level of social behavior, which is considered to be the pinnacle of social evolution**
9. The most familiar relative measures of water content of air is_____ **Relative humidity**
10. The rate at which organism can take in energy is___ either by external or internal constraints **Limited**
11. The earth ecosystem interacting with physical environment as a whole maintained a steady state system intermediate in the flow of energy between the high energy input of the sun and thermal sink of space is called_____ **Biosphere**
12. 1st multi-cellular organism found in China as carbonaceous filament were___ **1.8 Billion years**
13. Multicellular algae appear as carbonaceous filament from_____ old rock of china_**1.8 Billion years**
14. Human is in which order___ **Primate**
15. Maximum parsimony assumes that the _____(appearances of shared derived characters) is the most likely the **Tree that requires the fewest evolutionary events**
16. Oparin in 1924 gives_____ **Primeval soup theory**
17. In 18th Century, Carolus Linnaeus published a system of taxonomy on resemblance____**Two-part names of species and hierarchical classification**
18. One of the following is not example of horizontal gene transfer____ **Conjunction**
19. Which one of the following is not one of the positive interactions on microbe_____ **Amensalism**
20. Not Positive interaction _____ **Competition**
21. _____type of natural selection which favors more than two extreme phenotypes is **Disruptive selection**
22. Which one of the following is not one of the advantages of in situ conservation_____ **The Species have less space**

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23. The advantage of in-situ conservation except_____ **The specie will not evolve in their environment.**

Options 1-The species will have all the resources that it is adopted too

2-The species have more space

3-The species ll not evolve in their environment

24. Which of the following is not common extension of omics_____ **Vaccinomics**

25. Vertebrates make _____ % of total animals **5%**

26. Gymnopserms means_____ **Naked seeds**

27. Protozoa reproduce except_____ **Parthenogenesis**

28. Number of alleles in a sample population is called _____ **Allelic richness**

29. One of the first questions that is asked when a species is considered for whole-genome sequencing is **What is the size of its genome?**

30. Modern bread wheat is a mixture of recombined genes from _____ different wild species **3 Three**

31. Quinine is a drug used to treat_____**Malaria**

32. Annelids are not_____ **Predator**

33. Specie differ from foundation species is that they are likely to belong to_____ **higher trophic levels**

34. The_____ technique has now been replaced by sequencing using a mass spectrometry_____ **laborious edman degradation**

35. The first genome sequenced completely was that of_____**Homophilus influenza**

36. To assemble a genome from a series of sequences requires tremendous____ **Computational Power**

37. Community structure is measured by_____ **Species richness and diversity**

38. A resilient community or ecosystem may be _____ disrupted by disturbance but quickly return to former state_____ **Completely**

39. The ability to bounce back after disturbance is called_____ **Resilience**

40. _____ demonstrated that the influence of consumers on community structure depends upon their feeding preferences, their local population density, and the relative competitive abilities of prey species____ **Jane lubchenko**

41. Notochord is present throughout life or at least during embryonic stages in____ **Chordates**

42. The field of biology specially in developing hardware and software to store and analyze the huge amount of data generated by life science is called ___ **Bioinformatics**

43. Possibility of working with molecules on the scale of few micrometers is___ **Micro technology**

44. Phylogenetic Trees are evaluated in order to find the relationship among _____ **Species, Phylum, Family, (All of given)**

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45. Field experiment has_____ points **4 Four**

46. There are so-called _____that may be used to identify ecological important genes in incompletely characterized genome **Pre-genomic molecular approaches**

47. _____ are using these new tools to study large scale dynamic ecological phenomenon such as inter annual variation_____ **Ecologists**

48. The local population is density of the Glanville Fritillary_____ Melitaea cinxia, is lower on larger and on isolated habitat patches_____ **Butterfly**

49. _____cells are combination of proteins synthesizing system **Eukaryotic**

50. Sonoran desert plant distribution map clearly on the soils of different ages and form a_____ mosaic that closely matches soil mosaic **Vegetative**

51. _____ occur on nearly exposed geological substrate not significantly modified by organism **Primary Succession**

52. Supporting evidence for endosymbiosis except_____ Comes from studies of living **prokaryotic cells**

53. _____can be used to detect long-term ecological changes_____**Repeat Photography**

54. Who worked on the developing of paleontology___ **Smith, Buckland, Cuvier, Brongniart**

55. A food_____ summarizes the feeding relation in a community **Web**

56. _____ proposed the feeding activities of few species have inordinate influence on community structure _____ **Robert Paine**

57. Hardy-Weinberg principle conditions necessary to maintain constant alleles frequencies in a population except_____ **Immigration**

58. _____independently expanded the logistic model of population growth to represent interspecific competition **Lotka and Volterra**

59. _____explored the extent and sources of morphological variation in plant population including both the influence of environment and genetics **Clausen, Keck, and Hiesey**

60. The validity of field experiments depend upon several design features except_____ **Knowledge of final conditions**

61. Populations include_____ variations among individuals which are significant aspect of population structure **Genetic and Phenotypic effects**

62. _____soon found that even these so-called simple communities included very complex feeding relations **Charles Elton (1927)**

63. If for genetic modification, genes of the species or of a crossable plant are used under control of their native promoter, then they are called____ plants **Cisgenic**

64. A philosophical concept of value of something independent of its value to anything else is called_____ **Intrinsic Value**

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65. Mass spectroscopy is replaced by____ **Edman degradation**

66. Which of the following is an example of negative interaction_____**Amensalism**

67. The Nobel Prize winning scientist Linus Pauling Aptly described science as the____ **Search for Truth**

68. Study of transcripts that are present in an organism any time is called____ **Transcriptomics**

69. Mass spectrometry is used to determine the sequence of_____ **Amino acids**

70. Which of the following is not part of human genome project____**US department of medicine**

71. Comparative genomics is developing an increasing array of bioinformatics techniques. Which of the following is not done_____ **Synteny footprinting**

72. The bark of this tropical tree contains a drug called quinine is used in the prevention of_____ **Malaria**

73. Phylogenetic tree are evaluated in order to find the relationship among____ **Species, Phylum, Family, (All of given)**

74. What is not true about Protein degradation____ **Sequencing using mass spectrometry is replaced by admen degradation**

75. Medicines from nature account for usage by____ of the world's population___ **80%**

76. Habitat Destruction is major cause for_____ **Loss of Biodiversity**

77. Coral reefs which are biodiversity hotspots will disappear in ____year **20-40**

78. Plant founded in particular area is_____ **Flora**

79. Gregor mendel's Experiments with plant hybridization led to his establishing_____ **Laws of inheritance**

80. Genetic modification of plants is achieved by adding a specific gene or genes to a plant, or by knocking down a gene with_____ **RNAI**

81. The following are major activities of plant breeding; one of the following is the correct sequence of plant breeding_____ **Collection of variation-selection-evaluation-release(CSER)**

82. One of the following is the is not the correct sequence of deforestation_____ **Decrease in the level of carbon dioxide in the atmosphere**

83. One of the following is NOT the part of maintaining balance of the ecosystem____**Protection of food crops**

84. In Genome evolution Gene number & complexity of an organism____ **are not strongly linked**

85. Gene diversity is the variation at the level of___ **both at individual gene and in a population**

86. Haldane (1929) gave the _____**Hot, dilute soup Theory**

87. Not an example of Ex-situ Conservation_____ **Natural park and Natural Reserves**

88. One of the following a population over subsequent generations_____ **Bad or lethal mutations**

89. The fossil record is relatively incomplete for several reasons, which one is not true statement_____ **Soft tissues are mostly preserved**

90. Evolutionary fitness is a measure of_____ **Reproductive fitness**

91. _____uses DNA and other molecular data to determine evolutionary relationships **Molecular systematics**

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SUBJECTIVE

2 MARKS

Q-What are the processes of the ecosystem influenced by landscape structure?

Ans-Landscape structure influences processes such as the flow of energy, materials, and species between the ecosystems within a landscape.

Examples

Studies of the movements of small mammals in a prairie landscape show that a smaller proportion of individuals moves in more fragmented landscape but that the individuals that do move will move farther.

Q-Define Hardy-Weinberg principle.

Ans-One of the most fundamental concepts in population genetics, the Hardy-Weinberg principle, states that "in a population mating at random in the absence of evolutionary forces, allele frequencies will remain constant."

Q-Cuvier's advantage On natural selection?

Ans-Cuvier's anatomical and classificational work was impeccable.

- He identified four basic body plans for animals– vertebrates, molluscs, arthropods, and radiata (jellyfish & others).
- With these groups clearly distinguished, the chain of being was no longer part of biology.
- But links between distinct groups were still striking– Lamarck cited teeth in fetal baleen whales, for instance, which St. Hilaire took as an instance of 'unity of plan'. A broader sort of unity could restore the chain to some respectability…

3 Marks Questions

Q-How nutrient enrichment says the human is altering the aquatic and terrestrial ecosystem?

Nutrient enrichment by humans is altering aquatic and terrestrial ecosystems.

- ***Nitrate*** concentration and export by the earth's major rivers correlate directly with human population density.
- Human disturbance also increases export of ***phosphorus*** from aquatic catchments.
- Nutrient enrichment appears to be reducing the diversity of plants and fungi in terrestrial ecosystems.

Q-Define and explain optical foraging theory?

Optimal foraging theory attempts to model how organisms feed as an optimizin process.

Evolutionary ecologists predict that if organisms have limited access to energy, natural selection is likely to favor individuals that are more effective at acquiring energy and nutrients. Many animals select food in a way that appears to maximize the rate at which they capture energy.

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Plants appear to allocate energy to roots versus shoots in a way that increases their rate of intake of the resources that limit their growth. Plants in environments with abundant nutrients but little light tend to invest more energy in the growth of stems and leaves and less in roots. In environments rich in light but poor in nutrients, plants tend to invest more energy in the growth of roots.

Q-Lyell's Mistake on natural selection. ?

As a part of his general resistance to direction/ substantial change in geology, Lyell maintained there was no real direction of change in the fossil record either. This was greeted with pretty general astonishment, but Lyell claimed the apparent changes were just due to our having bad (misleading) samples…more mammals would soon appear in older formations, etc., as we extend our fossil collections. Within ten years (1840) it was clear that this was completely untenable.

Q-Proterozoic Animal Fossils ?

- Although scarce, a few animal fossils older than those of the Ediacaran fauna are known.
- A jellyfish-like impression is present in rocks 2000 m below the Ediacara Hills Pound Quartzite.
- Burrows in many areas presumably made by worms, occur in rocks at least 700 million years old.
- Wormlike and algae fossils come from 700 to 900 million-year-old rocks in China but the identity and age of these "fossils" has been questioned

Q-.How to documented genetic history in genome ?

Ans- An organism's evolutionary history is documented in its genome

- Comparing nucleic acids or other molecules to infer relatedness is a valuable tool for tracing organisms' evolutionary history
- DNA that codes for rRNA changes relatively slowly and is useful for investigating branching points hundreds of millions of years ago
- mtDNA evolves rapidly and can be used to explore recent evolutionary events

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5 Marks Questions

Q-Explain with example how exotic predators disturb the structure of food web. ?

Ans- Exotic predators and the structure of food webs

Exotic predators can collapse and simplify the structure of food webs. Introduced fishes have devastated the native fishes of Lake Atitlan and Gatun Lake in Central America.

Examples

The influence of the Nile perch on the fish community of Lake Victoria is enmeshed with massive changes in the lake's ecosystem. Introduction of the Nile perch is rapidly reducing the species-rich

fish fauna of Lake Victoria to a community dominated by a handful of species.

Humans

Humans have acted as keystone species in communities. People have long manipulated food webs both as a consequence of their own feeding activities and by introducing or deleting species from existing food webs. In addition, many of these manipulations have been focused on keystone species. Hunters in tropical rain forests have been responsible for removing keystone animal species from large areas of the rain forests of Central and South America. Chinese farmers have used ants as keystone predators to control pests in citrus orchards for over 1,700 years.

Q-Endosymbionts and evolution of eukaryotes?

Ans-

Formed from several prokaryotic cells . In a symbiotic relationship Symbiosis involving a prolonged association of two or more dissimilar organisms, is quite common today. In many cases both symbionts benefit from the association as occurs in lichens, once thought to be plants but actually symbiotic fungi and algae

Evidence for Endosymbiosis

Supporting evidence for endosymbiosis comes from studies of living eukaryotic cells containing internal structures called organelles, such as mitochondria and plastids which contain their own genetic material. In addition, prokaryotic cells synthesize proteins as a single system, whereas eukaryotic cells are a combination of protein-synthesizing systems.

Organelles Capable of Protein Synthesis

That is, some of the organelles within eukaryotic cells are capable of protein synthesis. These organelles with their own genetic material and protein-synthesizing capabilities are thought to have been free-living bacteria that entered into a symbiotic relationship, eventually giving rise to eukaryotic cells

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Q-Mutualistic Relationship?

Ans- Mutualism, interactions between individuals that benefit both partners, is a common phenomenon in nature that has apparently made important contributions to the evolutionary history of life and continues to make key contributions to the ecological integrity of the biosphere.

Mutualisms types

Mutualisms can be divided into those that arefacultative, where species can live without their mutualistic partners, and obligate, where species are so dependent on the mutualistic relationship that they cannot live without their mutualistic partners.

Examples

Plants benefit from mutualistie partnerships with a wide variety of bacteria, fungi, and animals.Mutualism provides benefits to plants ranging from nitrogen fixation and enhanced nutrient and water uptake to pollination and seed dispersal.

Mycorrhiza

Ninety percent of terrestrial plants form mutualistic relationships with mycorrhizal fungi, which make substantial contributions to plant performance. Mycorrhizae, which are mostly either vesicular-arbuscular mycorrhizae or ectomycorrhizae, are important in increasing plant access to water, nitrogen, phosphorus, and other nutrients.

In return for these nutrients, mycorrhizae receive energy-rich root exudates. Experiments have shown that the mutualistic balance sheet between plants and mycorrhizal fungi can be altered by the availability of nutrients. Plant-ant protection mutualisms are found in both tropical and temperate environments.

Ants

In tropical environments, many plants provide ants with food and shelter in exchange for protection from a variety of natural enemies. In temperate environments, mutualistic plants provide ants with food but not shelter in trade for protection.

Reef-building corals depend upon mutualistic relationships with algae and animals. The coral-centered mutualisms of tropical seas show striking parallels with terrestrial plantcentered mutualisms.

Mutualistic algae

Mutualistic algae called zooxanthellae provide reef-building corals with their principal energy source; in exchange for this energy, corals provide zooxanthellae with nutrients, especially nitrogen, a scarce resource in tropical seas.

Mutualism between corals

The mutualism between corals and zooxanthellae appears to be largely under the control of the coral partner, which chemically solicits the release of organic compounds from zooxanthellae and controls zooxanthellae population growth. Crabs and shrimp protect some coral species from coral predators in exchange for food and shelter.

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Q--Write significance of genetic life tree?

Investigating the Tree of Life

Phylogeny is the evolutionary history of a species or group of related species

The discipline of systematics classifies organisms and determines their evolutionary relationships

Systematists use fossil, molecular, and genetic data to infer evolutionary relationships

Phylogenies show evolutionary relationships

Taxonomy is the ordered division and naming of organisms

Phylogenetic trees do show patterns of descent
- Phylogenetic trees do not indicate when species evolved or how much genetic change occurred in a lineage
- it shouldn't be assumed that a taxon evolved from the taxon next to it
- Applications: whale meat sold illegally in Japan.

10 marks Questions

Q-How evolution of mutualism took place, how humans developed relationship with other species. ?

Evolution of Mutualism

Theory predicts that mutualism will evolve where the benefits of mutualism exceed the costs. **Keeler** built a cost-benefit model for the evolution and persistence of facultative plantant protection mutualisms in which the benefits of the mutualism to the plant are represented in terms of the proportion of the plant's energy budget that ants protect from damage by herbivores.

- The model assesses the costs of the mutualism to the plant in terms of the proportion of the plant's energy budget invested in extrafloral nectaries and the water, carbohydrates, and amino acids contained in the nectar. The model predicts that the mutualism will be favored where there are high densities of ants and potential herbivores and where the effectiveness of alternative defenses are low.

Humans in Mutualism

- Humans have developed a variety of mutualistic relationships with other species, but one of the most spectacular is that between the greater honeyguide and the traditional honey gatherers of Africa. In this apparently ancient mutualism, humans and honeyguides engage in elaborate communication and cooperation with clear benefit to both partners.
- The mutualism offers the human side a higher rate of discovery of bees' nests, while the honeyguide gains access to nests that it could not raid without human help. Careful observations have documented that the honeyguide informs the honey gatherers of the direction and distance to bees' nests as well as of their arrival at the nest.

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Q- Define ecology in our definition and write work of five different scientists.

Our Definition "Ecology is the scientific study of the processes regulating the distribution and abundance of organisms and the interactions among them, and the study of how these organisms in turn mediate the transport and transformation of energy and matter in the biosphere (i.e., the study of the design of ecosystem structure and function).

Beyond Fundamental Ecology Applied Ecology: Using ecological principles to maintain conditions necessary for the continuation of present day life on earth.

The work of different Scientists

1. **Kerner** showed that the differences in growth form observed by him and Bonnier were the result of temporary adjustments to environmental variation and not hereditary changes in the experimental plants.
2. **Turesson's** studies provided evidence for genetic differences among populations.
3. **Clausen, Keck, and Hiesey** explored the extent and sources of morphological variation in plant populations, including both the influences of environment and genetics.
4. **Case** determined that the best predictor of chuckwalla, Sauromalus, body length was average winter rainfall.
5. **Tracy's** laboratory growth experiments indicated that variation in body size among chuckwalla populations is at least partly determined by genetic differences among populations.
6. **Ledig and his colleagues** found a significant positive correlation between populafion size and genetic diversity in populations of Chihuahua spruce, a naturally fragmented population of trees living on mountain islands. **Frankham** showed that compared to mainland populations, island populations generally include less genetic variation.
7. **Saccheri and his colleagues** found that higher heterozygosity (genetic diversity) was associated with lower rates of population extinction through the effects
   of heterozygosity on larval survival, adult longevity, and egg hatching in populations of the Glanville fritillary butterfly, Melitaea cinxia.
8. **Studies by Carroll and several colleagues** show that soapberry bug populations living on native and introduced host plants have undergone natural selection for traits that favor their survival and reproduction on particular host plant species.

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Q: Evolutionary history of human?

Ans-HUMAN EVOLUTION

- Humans are too complex to be "understood" by any one field. Thus we will look at a
- Few major steps in evolution and some of the things affecting human evolution.
- Humans are members of the order Primates which consists of about 180 species (there
- Are 17 different orders of mammals which diverged 80-65 million years ago).
- Primates are a relatively old order of mammals. Most of the synapomorphies of this
- order are associated with an arboreal way of life: flexible digits, forward facing eyes,
- Vision as a primary sense.
- These traits may have played a role in the evolution of brain size in the lineage
- Leading to humans. Humans are a member of the family Hominidae which is believed
- to have diverged about 5 million years before the present (mybp) from the other
- Members of the Old world monkeys. At least 20 mybp the Hominoids split off from
- The other old world monkeys. The dates are rough and get changed now and then.

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BT201- ECOLOGY, BIODIVERSITY, EVOLUTION II

PAST-MCQS

1. Humans are included in the order ____**PRIMATES.**
2. The field of biology specializing in storing and analyzing data ____**BIOINFORMATICS**
3. The possibility of working with molecules on the scale of few micrometers is called. __**MICRO ECOLOGY**.
4. First question asked when a species is bought for genome sequencing ___**WHAT IS THE SIZE OF ITS GENOME?**
5. The study of transcripts that are present at any time in the cell are called ____**TRANSCRIPTOMICS**
6. Important measures used to describe community composition are___**SPECIES RICHNESS AND SPECIES DIVERSITY**
7. Vertebrates makes _____%of the animals.___**5%**
8. The number of allele in a sample ____ **ALLELIC RICHNESS**
9. Mass spectrometry used to determine ____sequence __**AMINO ACID SEQUENCE OF A PROTEIN**
10. Oparin (1924): work ___ **"PRIMEVAL SOUP" THEORY**
11. Quinine is obtained from quina it is used to treat ___**MALARIA**
12. The common extension of genomics are except____ **VACCINOMICS**
13. To assemble a genome from a series of sequence fragments requires tremendous __**COMPUTATIONAL POWER**
14. The first genome sequenced completely was that of___ **HEMOFELUES INFLUENZA**
15. The ___technique has now been replaced mostly by sequencing using mass spectrometry. **EDMAN DEGRADATION**
16. Annelids do not have feeding style ___**FILTER FEEDERS**
17. Modern bread wheat is a mixture of recombined genes from __**3**
18. Gymnosperms means ____**NAKED SEEDS**
19. Medicines from nature account for usage by ___of the world population.____ **80%**
20. Habitat destruction is the major cause ____ **BIODIVERSITY LOSS**
21. Coral reefs which are biodiversity hotspots will disappear in ____**20 TO 40 YEARS**
22. Plants found in a particular area is ____**FLORA**
23. Gregor Mendel experiments with plant hybridization led to this establishing of ____**LAWS OF INHERITANCE**

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24. Genetic modification of plant is achieved by adding a specific gene or genes to a plant or by knocking down a gene with to produce a desirable phenotype____. **RNAI**
25. They possess notochord in any period of their life ____**CHORDATA**
26. One of the following is not the consequences of deforestation ____ **DECREASE IN THE LEVEL OF CARBON DIOXIDE IN THE ATMOSPHERE**
27. If for genetic modification genes of the species or of a crossable plant are used under control of their native promoter, then they are called_____ Plants ____ **CISGENIC**
28. The following are the major activities of plant breeding. One of the following Gives the correct sequence of events in plant breeding ____ **COLLOECTION OF VARIATION- SELECTION -EVALUATION – RELEASE**
29. One of the following is the NOT part of maintaining balance of the ecosystem ____ **PROTECTING FOOD CROPS**
30. Which is not related to endosymbiosis **PROKARYOTIC CELL STUDY**
31. Which is not the part of horizontal gene transfer____ Conjugation • Transformation • Transduction **(OTHER THAN THIS)**
32. which One is not related to protozoa reproduction ____**PARTHENOGENESIS**
33. Taxonomy was established by ____**CAROLUS LINNAEUS**
34. Linus Pauling called science a way to find Truth or logic. select the correct one _____ **TRUTH**
35. ___Assumes that the tree that requires the fewest evolutionary events (appearances of shared derived characters) is the most likely_____ **MAXIMUM PARISMONY**
36. Annelids are not_____ **PREDATORS.**

1. The field of biology specializing in developing hardware and software to store and analyze the huge amount of data being generated by life scientists._____ **BIOINFORMATICS**
2. There are some so called ____ to identify ecologically genes in incompletely genome. ___**PRE- GENOMIC MOLECULAR APPROACHES).**
3. The international human genome project initiated by except.
   a) US national institutes of health
   b) US department of energy,
   c) Celera genomics ,
   d) **US NATIONAL INSTITUTES OF GENETICS**
4. Comparative genomics is developing an increasing array of bioinformatics techniques except.
   a) Synteny analysis
   b) phylogentic footprinting
   c) phylogenetic shadow,
   d) SYNTENY SHADOW
5. Keystone species differ from foundation specie, higher trophic level, and top predator; tend to modify their environment, **ALL GIVEN ABOVE**.

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6. Eukarya have been evolved form since ____ million years ago. ____**TWO (NOT CONFIRMED)**
7. Group consisting various species that lack a common ancestor called , _____**NONE**
8. type of selection which favor more than two extreme phenotype is _____**(DISRUPTIVE SELECTION)**
9. Which of the following is not mode of feeding of annelids_____ **PREDATOR**
10. A philosophical concept of value of something independent of its value to anything else called__ **(INTRINSIC VALU AND HERITBLE**
11. Krill Is a major food for_____ **WHALES**
12. Most important measures ecologists use to describe composition of community are___ **SPECIES RICHNESS AND SPECIES DIVERSITY**
13. Major technological developments of 1990s. , Computing, Communication , Micro technology , **ALL OF THE ABOVE**
14. ____are found in more than one copy in the genome. ____ **PARALOGOUS GENES**
15. Which is not true about protein degradation..
    a) Masses are estimated for peptide generated from proteolytic digest of protein
    b) fragment pattern obtained is compared entries in database that include sizes from genomic sequences
    c) Edman degradation technique is laborious
    d) Sequencing using mass spectrometry replaced by degradation

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SUBJECTIVE

2 Marks Questions

1. DEFINE COMMUNITY STRUCTURE?

Community structure is essentially the composition of a community, including the number of species in that community and their relative numbers. It can also be interpreted more broadly, to include all of the patterns of interaction between these different Species.

2. WHAT IS INTERMEDIATE DISTRUBANCE HYPOTHESIS?

The effects of human disturbance fall within the framework of the intermediate disturbance hypothesis. Though intense human disturbance reduces species diversity, moderate levels of disturbance may increase the diversity of some communities such as the European chalk grasslands

3. WHAT IS ECOLOGICAL GENOMICS?

A scientific discipline that studies the structure and functioning of a genome with the aim of understanding the relationship between the organism and its biotic and abiotic environments

4. ENLIST FOUR REASONS FOR LACKING OF FOSSIL RECORD?

1. Soft tissues are rarely preserved
2. Movement of the earth's crust has destroyed and covered many fossils
3. Fossilization takes place only in certain types of habitats and favorable environments
4. Paleontologists have not dug up every place on earth

5. DEFINE GENOME

A genome is an organism's complete set of DNA, including all of its genes. Each genome contains all of the information needed to build and maintain that organism.

6. DEFINE BIOINFORMATICS?

Bioinformatics is an interdisciplinary field that develops methods and software tools for understanding biological data. As an interdisciplinary field of science, bioinformatics combines biology, computer science, information engineering, mathematics and statistics to analyze and interpret biological data.

7. WHAT DOES INTERMEDIATE DISTURBANCE HYPOTHESIS SUGGEST?

The intermediate disturbance hypothesis suggests that communities with a medium (intermediate) level of disturbance may have greater species diversity than communities with very frequent or very rare disturbances.

8. ROLE OF PRIMERS IN PCR?

A primer is a short single strand of RNA or DNA (generally about 18-22 bases) that serves as a starting point for DNA synthesis. It is required for DNA replication because the enzymes that catalyze this process, DNA polymerases, can only add new nucleotides to an existing strand of DNA in the PCR.

9. PROTEOMICS DEFINITION?

Proteomics is the study of all the proteins in the cell. Mass spectrometry may also be used to determine the amino acid sequence of a protein. For this application, the protein is cleaved with a protease.

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10. ENLIST TYPES OF SERVERS USED IN BIOINFORMATICS?

1. WWW servers
2. Database servers
3. Intensive computing servers

11. ENLIST THREE MAJOR TECHNOLOGICAL DEVELOPMENTS OF 1990S?

1. Micro technology
2. Computing
3. Communication

12. HOW SEQUENCE ASSEMBLY IS DONE USING SHOTGUN APPROACH?

Master sequence ← short sequences, simply by examining the sequences for overlaps. No need any prior knowledge of the genome

13. WHAT ARE THREE LAYERS OF GENOME ANNOTATION?

1. Nucleotide-level annotation
2. Protein level annotation
3. Process level annotation

14. DIFFERENTIATE BETWEEN SPECIES TREE AND GENE TREE.

SPECIES TREE	GENE TREE
Contains only one representative from each species	Often contains a number of genes from a single species
All nodes indicate speciation events	Nodes relate either to speciation or gene duplication events.

15. DEFINE MICRO ECOLOGY?

The possibility of working with molecules on the scale of a few micrometers, given by advances in laser technology, has been very important for one of genomics' most conspicuous achievements, the development of the gene chip.

16. WHAT IS THE HIDDEN MARKOV MODEL

It may generally be used in pattern recognition problems, anywhere there may be a model producing a sequence of observations. In bioinformatics, it has been used in sequence alignment, in silico gene detection, structure prediction, data-mining literature, and so on.

3 MARKS QUESTIONS

1. EXPLAIN SPECIES RICHNESS AND SPECIES DIVERSITY WITH EXAMPLE?

SPECIES RICHNESS is the number of different species in a particular community. **For example** if we found 30 species in one community, and 300 species in another, the second community would have much higher species richness than the first. Communities with the highest species richness tend to be found in areas near the equator and Communities with the lowest species richness lie near the poles

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SPECIES DIVERSITY is a measure of community complexity. It is a function of both the number of different species in the community (species richness) and their relative abundances (species evenness. **For example:** • A forest community with 20 different kinds of trees would have greater species diversity than a forest community with only 5 kinds of trees.

2. HOW THE EVOLUTIONARY HISTORY CONVERTS IN TO DOCUMENT?

- Comparing nucleic acids or other molecules to infer relatedness is a valuable tool for tracing organisms' evolutionary history
- DNA that codes for rRNA changes relatively slowly and is useful for investigating branching points hundreds of millions of years ago
- mtDNA evolves rapidly and can be used to explore recent evolutionary events

3. NAME THE PHYLA IN AMOEBOID PROTOZOANS? WRITE THE NAME OF THE PHYLA WHICH INCLUDES AMOEBOID PROTOZOANS AND DESCRIBE THEIR LOCOMOTION?

Phylum: - Sarcodinids (Amoeboid protozoan)

Contains 4 phyla: Foraminifera, Radiozoa, Amoebozoa, and Heliozoa

Locomotion:- locomotion occurs due to the streaming of the cytoplasm, that results into the formation of the pseudopodia "False feet", and it helps amoeba for the locomotion.

Other Characteristics

- Most reproduce asexually through binary fission.
- Food is usually captured by phagocytosis.
- Body types range from free flowing to rigid with skeletal supports

4. INTERACTION B/W GENES WITHIN GENOME?

The interactions between the genes within the genome and the dynamic character of the genome on an evolutionary scale have been sketched brilliantly by Dover (1999) as an internal tangled bank. This idea goes back to Darwin (1859) who, after investigating the banks of hollow roads in the English countryside, was attracted by the great variety of organisms tangled together.

Darwin considered the way in which all organisms depended on each other as the template for evolution. Inspired by Darwin, Dover (1999) made a distinction between the 'external tangled bank' (the ecology) and the 'internal tangled bank' (the genome), attributing to them complementary roles in the evolutionary process

5. HOW KEYSTONE SPECIES DIFFER FROM FOUNDATION SPECIES?

- A keystone species is a species that has a extremely large effect on community structure relative to its biomass or abundance.
- Keystone species differ from foundation species in two main ways
  i. they are more likely to belong to higher trophic levels (to be top predators),
  ii. they act in more diverse ways than foundation species, which tend to modify their environment

6. EXPLAIN SEQUENCING?

Sequencing means to make DNA/Protein short fragments into in an order.

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Protein sequencing

The laborious edman degradation technique has now been replaced mostly by sequencing using mass spectrometry, in which masses are estimated for peptides generated from proteolytic digests of the protein, while the fragment pattern obtained is compared with entries in a database that include sizes predicted from the genomic sequence.

7. WRITE AT LEAST THREE REASONS TO USE PHYLOGENETIC.

1. **Understand the lineage of different species:**
2. **Understand how various functions evolved:**
3. **Perform multiple sequence alignment**
4. **Predict gene function (phylogenetic footprint)**

5 MARKS Questions

1. EXPLAIN IMPORTANCE OF BIODIVERSITY?

Biodiversity has a number of functions on the Earth.

These are as follows:

- Maintaining balance of the ecosystem:
- Recycling and storage of nutrients,
- combating pollution, and stabilizing climate, protecting water resources,
- Forming and protecting soil and maintaining Eco-balance.
- Intrinsic value
- A philosophical concept of the value of something independent of its value to anyone or anything

Provision of biological resources

- Provision of medicines and pharmaceuticals, food for the human population and animals, ornamental plants, wood products, breeding stock and diversity of species, ecosystems and genes.
- Quinone is a drug used for the treatment of malaria which is obtained from the Cinchona genus of plants.

Social benefits:- Recreation and tourism, cultural value and education and research. The role of biodiversity in the following areas will help make clear the importance of biodiversity in human life:

Biodiversity and food: - 80% of human food supply comes from 20 kinds of plants. But humans use 40,000 species for food, clothing and shelter. Biodiversity provides for variety of foods for the planet.

Biodiversity and human health:- The shortage of drinking water is expected to create a major global crisis. Biodiversity also plays an important role in drug discovery and medicinal resources. Medicines from nature account for usage by 80% of the world's population.

Biodiversity and industry:- Biological sources provide many industrial materials. These include fiber, oil, dyes, rubber, water, timber, paper and food.

Biodiversity and culture: - Biodiversity enhances recreational activities like bird watching, fishing, trekking etc. It inspires musicians and artists.

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2. WHAT IS EVOLUTIONARY SIGNIFICANCE OF SEQUENCING GENOME IN ECOLOGY; EXPLAIN WITH EXAMPLES.

Ecological significance

It will be clear that ecological arguments have only played a minor role in the selection of species for whole-genome sequencing, but we expect them to become more important in the future. The new range of models should embrace diverse phylogenetic lineages, varying in their physiology and life-history strategy. **For example**, the model plants arabidopsis and rice both employ the C3 photosynthetic pathway. To complement our genomic knowledge of primary production, new models should be chosen among plants utilizing C4 photosynthesis or crassulacean acid metabolism (CAM). Considering the diversity of life histories, species differing in their mode of reproduction and dispersal capacity should be chosen; **for example**, hermaphoditism versus gonochorism, parthenogenesis versus bisexual reproduction, etc

3. What are phylogenetic trees (how genes are related)

- Phylogentics tree or evolutionary tree is a diagram or branching tree.
- Which shows the evolutionary relationship between the different species depending upon their phylogeny based upon similarities and differences in their physical and genetic characteristics
- All the life on earth is part of the single phylogenic tree showing only one common ancestor and all species have been evolved from that common ancestor.

Reasons for using the phylogenetic trees

- Understanding the linage of the different species
- Understanding how the various functions evolved among different species
- Perform multiple sequence alignment
- Predict gene functions (phylogenetic footprints)

How genes are related

- Gene tree contains no of genes from a single species
- Every node of the gene tree shows the speciation or gene duplications events.
- Gene families are groups of related genes that share a common ancestor.
- Paralogs are genes with similar sequences from within the same species
- orthologs are genes with similar sequences in different species

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4. WRITE FIVE FACTORS THAT AFFECTS COMMUNITY STRUCTURE ?

The structure of a community is the result of many interacting factors, both abiotic (non-living) and biotic (living organism-related).

Here are some important factors that influence community structure:

- The climate patterns of the community's location.
- The geography of the community's location.
- The heterogeneity (patchiness) of the environment
- The frequency of disturbances, or disruptive events.
- Interactions between organisms.

5. WHY THE STUDY OF WHOLE GENOME IS IMPORTANT FOR THE EVOLUTIONARY STUDY?

- Whole-genome analysis of organisms at crucial or disputed positions in the tree of life can be expected to contribute significantly to our knowledge of evolution.
- The sea squirt, **ci. Intestinalis**, was chosen as a model because it belongs to a group, the Urochordata, with properties similar to the ancestors of vertebrates.
- The study of this species should provide valuable information about the early evolution of the phylum to which we belong ourselves.
- Many other organisms, although not on the list for a genome project to date, have a strong case for being declared as model species for evolutionary arguments. These include the velvet worm, Peripatus, traditionally seen as a missing link between the arthropods and annelids, but now classified as a separate phylum in the panarthropoda lineage , and the springtail, folsomia candida, formerly regarded as a primitive insect, but now suggested to have developed the hexapod body plann before the insects separated from the crustaceans.

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